

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B. Pharm. Examination
University Theory Examination September 2018
BPH2002 Pharmaceutical Organic Chemistry-I

Date: 17.09.2018, Monday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 75

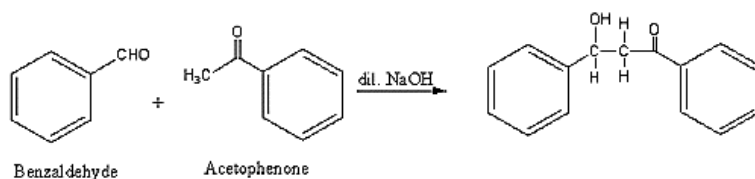
Instructions:

- [A] There are three sections in this question paper.
- [B] SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
- [C] Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
- [D] Figures to right indicate marks.
- [E] Draw neat labeled sketches wherever necessary.

SECTION - I

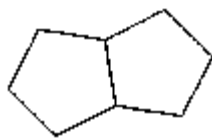
Q 1 Attempt all questions. Each question is of one mark. 20

1. The Saytzeff's rule is
 - [A] It form more stable alkene with greater no. of alkyl group
 - [B] It form more stable alkene with less no. of alkyl group
 - [C] It form least stable alkene with greater no. of alkyl group
 - [D] It form least stable alkene with less no. of alkyl group
2. For which mechanisms the first step involved is the same?
 - [A] E₁ and E₂
 - [B] E₂ and S_N2
 - [C] E₁ and S_N1
 - [D] S_N1 and S_N2
3. Identify the following reaction;



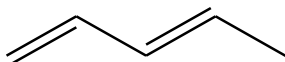
- [A] Aldol Condensation
 - [B] Wolf Kishner Reduction
 - [C] Benzoin condensation
 - [D] Perkin condensation
4. Which one of the following acids is present in lemon juice?
 - [A] Citric acid
 - [B] Benzoic acid
 - [C] Tartaric acid
 - [D] Oxalic acid

5. Which of the following is the reagent for preparation of alkyl halide from alkene
 - [A] SOCl_2
 - [B] PBr_3
 - [C] Cu in ether
 - [D] HBr
6. Which of the following statements about amines is incorrect?
 - [A] They react with acids to form salts.
 - [B] Aliphatic amines are more basic than aromatic amines.
 - [C] Primary amines are more basic than tertiary amines.
 - [D] Aniline is Primary aromatic compound
7. Cannizzaro reaction is not given by
 - [A] Formaldehyde
 - [B] Acetaldehyde
 - [C] Benzaldehyde
 - [D] Trimethyl acetaldehyde
8. Which of the following statements is not correct?
 - [A] Aldehydes and ketones undergo nucleophilic addition
 - [B] Aldehydes and ketones undergo electrophilic substitution
 - [C] Aldehydes and ketones contain polar carbonyl group
 - [D] Lower members of aldehydes and ketones are soluble in water due to hydrogen bonding
9. Lucas test is used to differentiate _____
 - [A] Alcohol
 - [B] Amine
 - [C] Phenol
 - [D] Anilide
10. Identify the following structure;

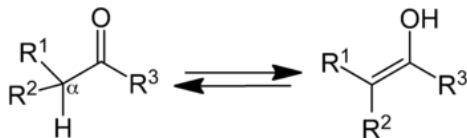


- [A] Spiro
 - [B] Fused
 - [C] Bridgehead
 - [D] Heterocyclic
11. Which is the example of Homogenous Catalyst?
 - [A] Pd
 - [B] Ni
 - [C] Lindler
 - [D] Wilkinson's catalyst
 12. The $\text{S}_\text{N}1$ reaction is reaction of alkyl halide with nucleophile in which the product configuration is
 - [A] Only Walden inversion
 - [B] Only Retention configuration
 - [C] Both Walden and Retention of Configuration
 - [D] None of Above

13. Identify the following type's dienes.



- [A] Conjugated Diene
 - [B] Isolated Diene
 - [C] Cumulated Diene
 - [D] None of above
14. Which of the following is more acidic?
- [A] Acetic acid
 - [B] Trifluoroacetic acid
 - [C] Fluoroacetic acid
 - [D] Difluoroacetic acid
15. The IUPAC name of isopropyl alcohol is
- [A] 2-Propanol
 - [B] 1-Propanol
 - [C] 2-Propanal
 - [D] 1-Propanal
16. Which molecule contains a Sp^2 -hybridized atom?
- [A] Water
 - [B] Ethylene
 - [C] Acetylene
 - [D] Ammonia
17. Orientation of Elimination reaction follows...
- [A] Markoniov's rule
 - [B] Saytzeff rule
 - [C] Micheal addition
 - [D] Hoffmann rule
18. Addition of water to alkene will give_____.
- [A] Aldehyde
 - [B] Ketone
 - [C] Alcohol
 - [D] Carboxylic acid
19. Saturated hydrocarbons are also referred as
- [A] Alkanes
 - [B] Alkenes
 - [C] Alkynes
 - [D] Alkaloids
20. Identify the following structural isomer.



- [A] Chain Isomer
- [B] Tautomer's
- [C] Metamerism
- [D] Functional Isomer

SECTION – II

- Q 2** Explain reaction, mechanism, reactivity, kinetic and stereochemistry of SN_1 reaction. Write various factors affecting rate of SN_1 reaction. (10)
- Q 3 Attempt any TWO of the following** (10)
- A** Write a note on structural isomerism in organic compounds. 05
- B** Explain reaction, mechanism, reactivity, kinetic and orientation of E_2 reaction. 05
- C** Write structure of following IUPAC name. 05
- i. 2,3 Dimethyl Butane
 - ii. 2- Butene
 - iii. 2-Butanone
 - iv. Butenoic acid
 - v. 3-Chloro pentanol

SECTION – III

- Q 4 Attempt any ONE of the following** (10)
- A** Write any three preparation and reaction of aldehydes and ketones. 10
- B** Write reactions alkane and alkene and explain free radical mechanism for halogenation of alkane. 10
- Q 5 Attempt any FIVE of the following** (25)
- A** Write any three preparation and reaction of alcohol. 05
- B** Explain 1, 4 addition Vs. 1, 2 addition in dienes. 05
- C** Write a short note on Cannizzaro reaction and Aldol Condensation 05
- D** Explain Hofmann degradation of amides with suitable reaction and mechanism. 05
- E** Write Markovnikov's and Saytzeff's rules with suitable examples. 05
- F** Write structure and use of following: (Any Five) 05
- i. Methyl salicylate
 - ii. Vanillin
 - iii. Propylene glycol
 - iv. Amphetamine
 - v. Benzyl alcohol
 - vi. Chloroform
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